

NEW AOI TRENDS AND THEIR BENEFITS

Trend	Detail	Innovation
AI and machine learning integration	Leveraging AI/ML for defect classification and adaptive learning to reduce false positives	Systems can learn from operator feedback, improving accuracy over time
3D imaging and inspection	Use 3D imaging to detect solder joint defects and height measurements accurately	3D AOI can inspect components from multiple angles, providing a holistic view. Examples are Koh Young Zenith UHS, CyberOptics SQ3000 and SAKI 3Di Series
High-resolution cameras	Cameras with resolutions up to 25 MP or higher can be deployed for precise inspection	Enhanced image quality and detailed defect detection. Examples Mirtec MV-7 Omni and PARMi SIGMA X Series
Deep learning algorithms	Advanced algorithms for feature recognition and anomaly detection	Ability to detect subtle or non-standard defects. Example: Omron VT-S10, ViTrox V310i
Edge computing	On-device data processing for real-time inspection	Reduces latency and improves system efficiency
Connectivity and IIoT (industrial internet of things)	Integration with factory networks for real-time monitoring and analytics	Enables predictive maintenance and smart manufacturing
Large-format PCB support	AOI systems are capable of inspecting oversized PCBs	Accommodates advanced designs and larger assemblies. Example: Test Research, Inc. TR7500QE
Dual-lighting technology	Use multiple lighting sources to enhance defect visibility	Improves contrast and identification of subtle defects
Software-defined AOI	Flexible software updates to adapt to evolving manufacturing needs	Extends system lifespan and adapts to new component designs
High-speed inspection	Systems optimised for fast throughput without compromising accuracy	Critical for high-volume manufacturing environments. Nordson YESTECH FX-942

or insufficient solder in BGAs and other hidden joints

Internal component damage.

Issues within components that might not be externally visible

Multi-layer PCBs. Manufacturing flaws within stacked layers of a PCB

Using either 2D or 3D imaging techniques, X-ray systems provide unparalleled visibility into the subsurface structures of PCBAs. Although slower and more expensive than optical AOI, they are indispensable for fault-critical sectors like aerospace and medical devices, where fault tolerance is minimal.

Laser AOI: Precision meets depth measurement

Laser-based AOI systems represent a newer addition to the inspection toolbox, offering capabilities that complement optical and X-ray systems. By using laser triangulation or interferometry, these systems:

Measure height and depth. Precisely quantify solder paste volume to ensure proper deposition before reflow

Inspect non-reflective surfaces.

Overcome challenges posed by materials that absorb or scatter visible light

Enhance accuracy. Provide micron-level measurements for critical components

Laser AOI is particularly valuable in solder paste inspection (SPI) and for validating the coplanarity of components before soldering. While not as widely adopted as optical or X-ray systems, lasers add a layer of precision that is invaluable in specific use cases.

As PCB designs grow increasingly intricate and component densities rise, AOI systems are essential for maintaining product quality. Optical and X-ray technologies form the backbone of AOI, while laser systems carve out a niche in precision-demanding applications.

Many manufacturers integrate multiple AOI technologies in their production lines to achieve the best results. For example:

- Optical AOI is used for initial inspections to catch visible defects.
- Laser AOI ensures accurate solder paste deposition.
- X-ray AOI verifies the integrity

of hidden solder joints and internal structures.

By leveraging the strengths of each technology, manufacturers can ensure comprehensive quality assurance, reduce rework costs, and maintain high standards for their products.

Key innovations in AOI systems

Inline automated optical inspection.

Inline AOI systems are seamlessly integrated into automated manufacturing lines, ensuring continuous PCB inspection without manual intervention. This integration enables manufacturers to identify and address defects immediately, improving efficiency and product quality. For instance, Mirtec's MV-6 and Panasonic's NPM-W2 models offer robust inline AOI capabilities.

Kumud Tyagi, CEO of Bergen Associates Pvt Ltd, highlights, 'The Evolution of In-Line AOI: Transforming PCB Manufacturing with AI and Speed.' She mentions that inline AOI systems, equipped with AI capabilities, are rapidly becoming the cornerstone of efficient production lines.